Exploring the State-of-Receptivity for mHealth Interventions: Supplementary Material

FLORIAN KÜNZLER*, ETH Zürich
VARUN MISHRA*†, Dartmouth College
JAN-NIKLAS KRAMER, University of St. Gallen
DAVID KOTZ, Dartmouth College
ELGAR FLEISCH, ETH Zürich and University of St. Gallen
TOBIAS KOWATSCH, ETH Zürich and University of St. Gallen

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S1 EXPECTATION OF DIRECTION OF EFFECTS FROM PRIOR WORK

In this work we are exploring associations from the data collected and participants' receptivity to intervention messages. This analysis for interventions has not been explored before and hence is one of our primary research questions. In the past, researchers have observed associations between different factors and response to notifications or EMA and have observed similar and sometimes conflicting trends. Given that there is no clear consensus for the observed trends in past work – maybe because of the population size, or population demographics or device type or different measure of receptivity/availability – we chose to conduct an exploratory analysis on a representative population using different devices.

However, based on list our expectation of the direction of effects based on observations from previous work. We formulate our hypotheses on the entire receptivity metric: response rate, conversation rate and response delay. Hence higher receptivity would mean higher response rate, higher conversation rate and lower response delay.

*Both authors contributed equally to this research.
†Corresponding author

Authors' addresses: Florian Künzler, ETH Zürich; Varun Mishra, Dartmouth College, varun@cs.dartmouth.edu; Jan-Niklas Kramer, University of St. Gallen; David Kotz, Dartmouth College; Elgar Fleisch, ETH Zürich, University of St. Gallen; Tobias Kowatsch, ETH Zürich, University of St. Gallen.

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- Age: Pielot et al. observed that age had a positive correlation with engagement with notifications [3]. Hence, we hypothesize older participants would be more likely to be receptive as compared to younger participants.
- Gender: In the same study, Pielot et al. did not observe any association between gender of the user and engagement with notifications. Hence, we hypothesize gender would not have a significant effect on receptivity.
- Personality: Mehrotra et al. observed that extroversion and neuroticism affect how fast people respond to notifications [2]. Higher extroversion led to longer response time, and higher neuroticism led to faster response time. Hence, we hypothesize these traits should influence receptivity. Higher neuroticism should lead to higher receptivity and higher extroversion should lead to lower receptivity.
- Time of day: There is no conclusive evidence if time of day has a significant effect on receptivity. Westermann et al. found that replying solely on the time of the day for sending notifications had no impact on how fast people engaged with the content [6]. However, Pielot et al. found that notifications posted later in the day had higher engagement rates [3]. In fact, they found that engagement levels remained similar throughout the day and only lowered during early hours of the day (7-8 am). Based on observation by Pielot et al. we hypothesize that receptivity towards interventions would be lower during the early hours of the day, as compared to the rest of the day.
- Phone Battery: Pielot et al. found a weak positive correlation between phone battery level and engagement rates ($r = 0.008, p < 0.05$) [3]. They concluded that participants were more likely to engage if the battery level was higher. However, given the low correlation coefficient (r), we do not expect to observe any significant effect of battery level and receptivity metrics.
- Device Interaction: Pielot et al. found that number of phone unlocks correlate with increased feelings of boredom, which in turn correlated with openness to consume entertaining news articles (i.e., engaging the user) [4]. In a different work, Pielot et al. found that participants were more likely to engage with notifications the less time had passed since their last device interaction (change of screen status) [3]. We formulate our hypotheses for receptivity based on the observation by these works.
- Physical Activity: In previous work, Sarker et al. observed that walking did not indicate availability towards EMA prompts [5]. However, Pielot et al. found that more difficult to predict activities (such as driving, cycling, walking) correlate with higher openness for engagement, as compared to being still [3]. Mehrotra et al. found that certain activities (like cycling) were less opportune moments for delivering notifications, as compared to other activities (like walking). Given the lack of clear consensus on effect of physical activity, we hypothesize that walking would relate with higher receptivity as compared to other activities.
- Location: Similar to physical activity, there is no clear consensus about effect of location. Mehrotra et al. found that type of location (home, work, other) did not have a significant effect on response times to notifications [1]. However, Sarker et al. found that, as compared to other locations, participants at work were less available to respond to EMA [5]. Hence, we hypothesize that participants at work will be less receptive.

S2 CORRELATION ANALYSIS OF FACTORS

There could be correlations and interdependencies amongst the different factors we considered in our work. For the intrinsic factors, we believe, such correlations (if any) could be grossly misleading about the inference we make. For example, if gender and device type is strongly correlated and we make an inference that women are more receptive, it could be because of a particular device type. In this case, device type could have *caused* the increase receptivity, and not because women are more receptive. To this end, we perform a correlation analysis

between the different intrinsic factors; the result is shown in Fig. 1. We did not observe several strong correlations except between gender and age. Women negatively correlated with age, implying in our study, women were younger as compared to men. However, this did not affect receptivity, as we observed that age had an effect on receptivity metrics, but gender did not.

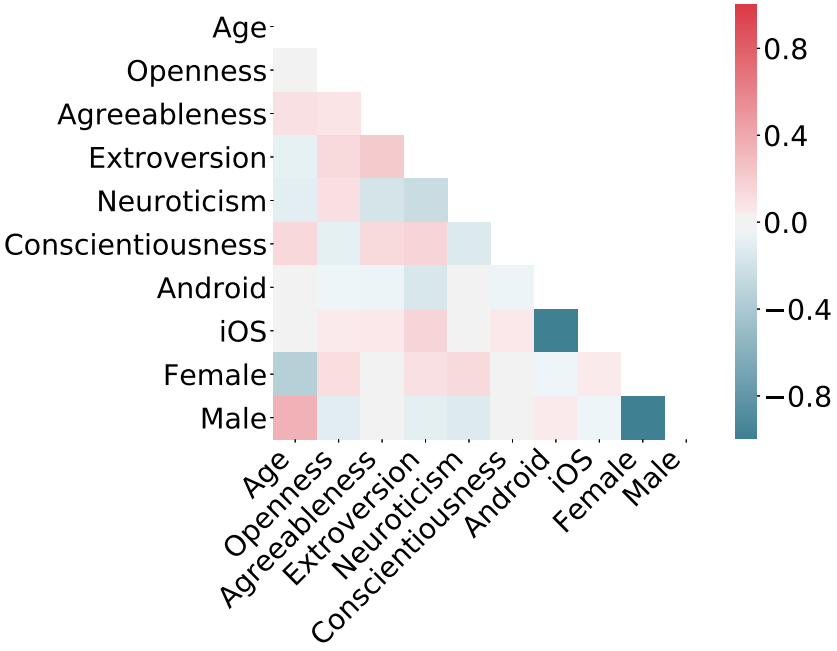


Fig. 1. Correlation analysis of the different intrinsic factors

However, if correlations exist between the contextual factors, it is not a major problem. We argue that all the contextual factors that are sensed from the smartphone are independent of each other, i.e., the inference from the analysis of one sensor would not change regardless of the presence of another 'correlated'. This is because if a sensor signal is associated with a behavioral pattern (in our case, receptivity), there was a 'cause' that led to that behavior. We simply capture an effect of that 'cause' in our various sensor signals. In no circumstance do we imply that the signal caused or led to a particular behavior. As a naive example, let us *assume* people charge their phone when working in office, and because they are working they are less receptive. So here the factor that is 'causing' them to be less receptive is the fact that they are working. Signals like phone battery state, work location and time of day are just signals affected by the 'cause'. In this case, the three signals are correlated, but we cannot just discard two of them because we might think one of the signals caused the behavior. We believe, all three signals are necessary to best capture the 'cause'. This is one of the major reason we considered each factor individually, instead of combining them together in a multi-variate model, where the effect of a correlated signal might be minimized, i.e., a correlated variable might not be significant in a multi-variate model, but significant in a bivariate model.

However, for completeness, we perform correlation analysis on the different contextual factors for iOS and Android users separately, and show the results in Fig. 2 & 3, respectively. We observed several interesting characteristics, e.g., WiFi connected is strongly correlated to users being at home. This is one example of our

reasoning to evaluate each factor individually. Assuming, researchers did not have location information, they could monitor Wi-Fi state to make meaningful inferences. Similarly, we observe that battery full is correlated with time since last screen state change (or lock/unlock time). This suggests that if the battery is full, the phone has been charging for a while and not in use by the participants, thereby making them less receptive, consistent to our analysis in the paper. This is another example, where analyzing the factors individually provides useful information for detecting receptivity.

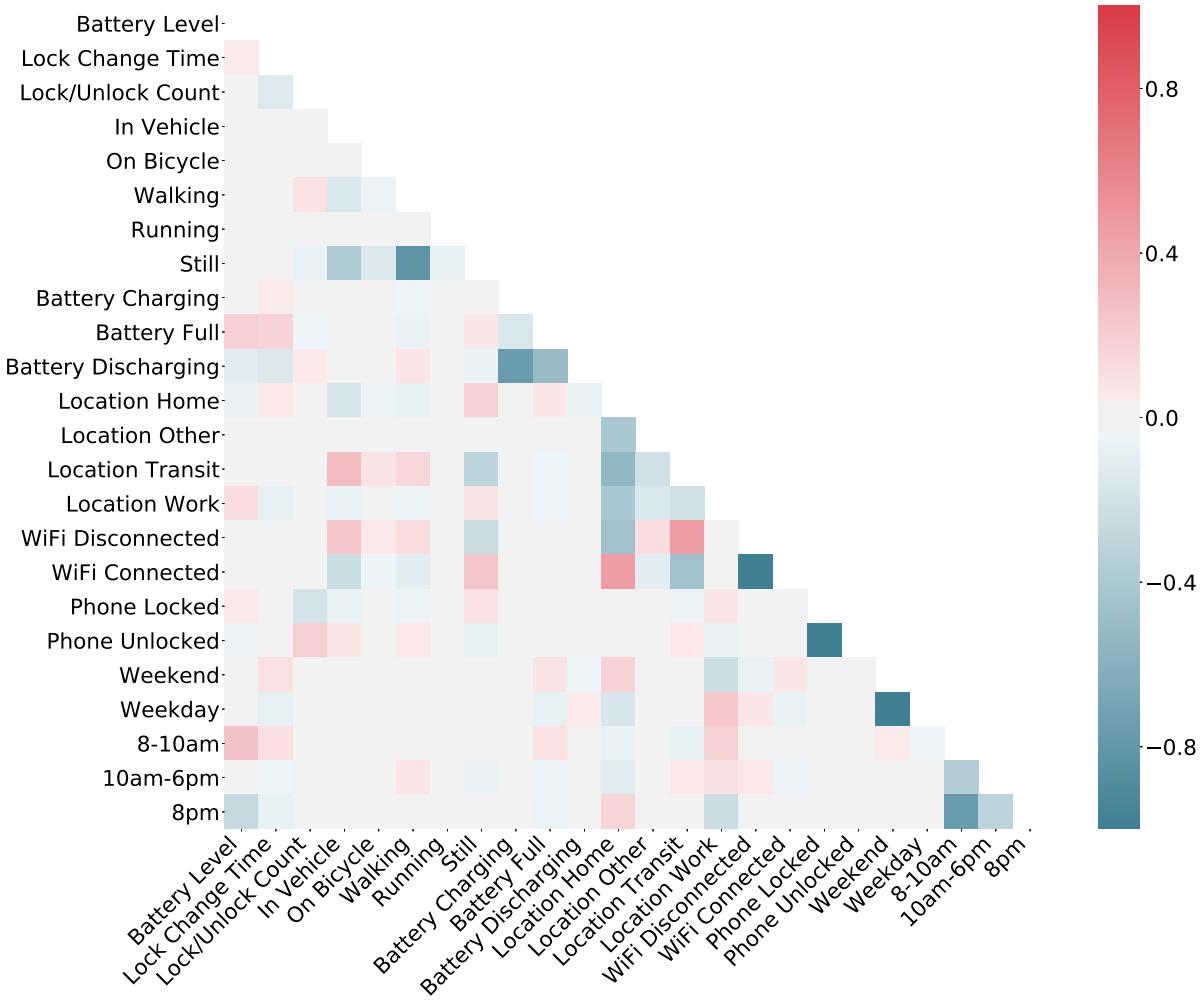


Fig. 2. iOS: Correlation analysis of the different contextual factors

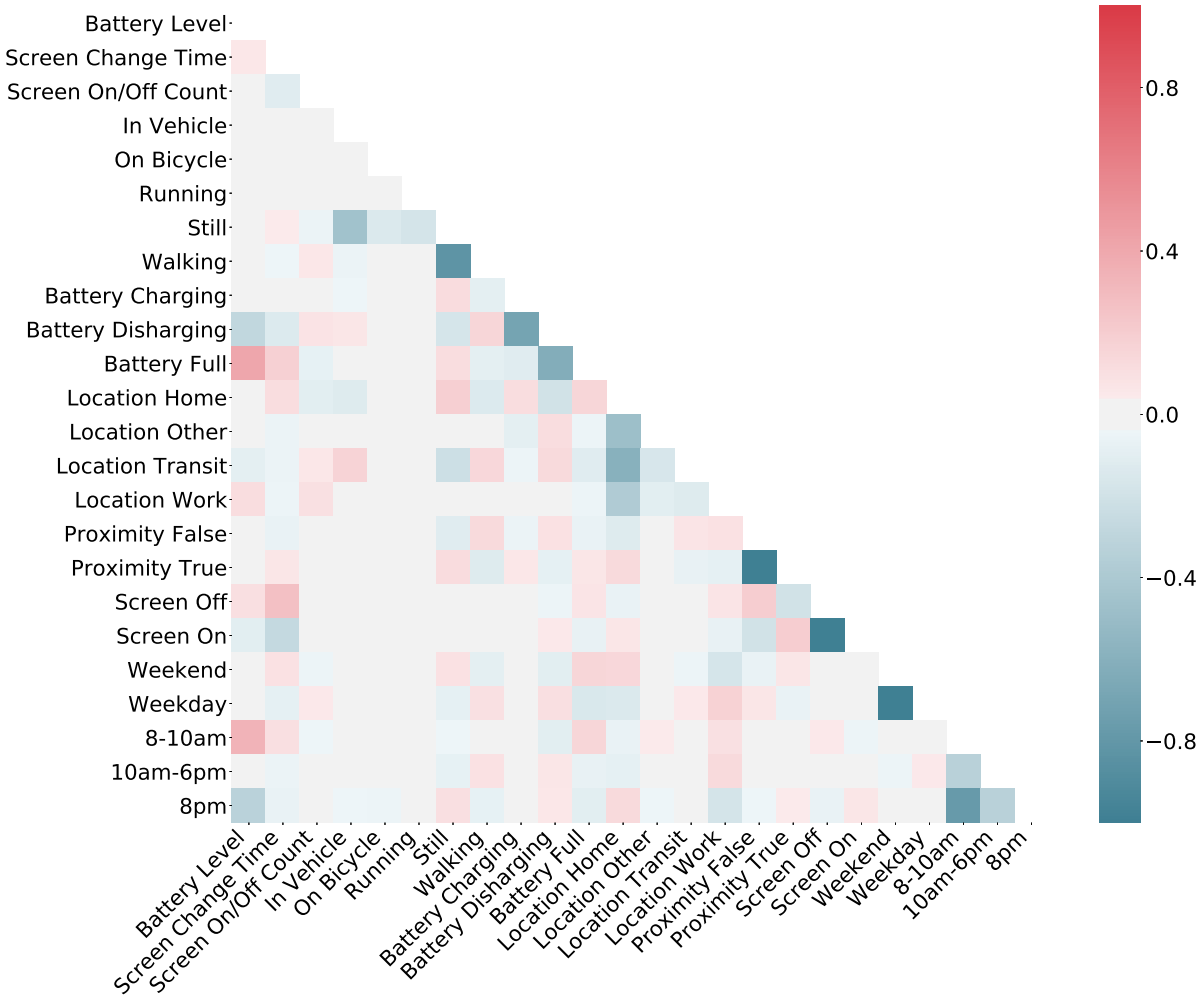


Fig. 3. Android: Correlation analysis of the different contextual factors

Table 1. **iOS**: Coefficients from a multi-linear regression model to understand to understand the influence of the different contextual factors and the different receptivity metrics

	Coefficients					
	Just-in-time response		*Conversation engagement*		*Time delay*	
(Intercept)	0.245	***	0.190	***	120.827	***
Activity (in a vehicle)	0.018		0.014		−10.339	
Activity (on a bicycle)	−0.009		−0.002		−9.628	
Activity (on foot)	0.045	***	0.040	***	−15.861	**
Activity (running)	0.020		−0.026		13.877	
Battery level	0.001		0.000		3.134	
Battery status (charging)	0.012		0.014		−0.764	
Battery status (full)	−0.039	**	−0.031	*	10.568	
Time since last lock	−0.011	**	−0.007		3.581	
Lock state (locked)	−0.027	**	−0.025	**	−2.909	
Location (Other)	−0.027	*	−0.018		5.484	
Location (Transit)	−0.008		−0.003		−0.773	
Location (Work)	−0.015		−0.015		19.214	*
Unlock count	0.013	**	0.006		−2.800	
Wi-Fi (not connected)	−0.034	***	−0.026	**	8.890	
Weekend	−0.027	**	−0.027	**	11.376	*
Morning message (8-10am)	−0.015		−0.002		26.884	***
Afternoon message (10am-6pm)	0.078	***	0.013		−25.834	**

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

S3 MULTI-LINEAR REGRESSION ANALYSIS

In this section we perform multi-linear regression analysis to understand which *contextual* factors are significant in a more controlled sense. Since we had different sensors for iOS and Android users, we performed the multi-linear regression analysis for the two device types separately. Table 1 shows the coefficients for the contextual factors with the different receptivity metrics (just-in-time response, conversation engagement and response delay). Table 2 shows the coefficients for the Android users.

Table 2. **Android**: Coefficients from a multi-linear regression model to understand to understand the influence of the different contextual factors and the different receptivity metrics

	Coefficients					
	Just-in-time response		*Conversation engagement*		*Time delay*	
(Intercept)	0.127	*	0.080		135.176	***
Activity (in a vehicle)	−0.009		−0.033		17.630	
Activity (on a bicycle)	0.140		0.038		−75.178	
Activity (running)	−0.043		−0.114		−13.858	
Activity (walking)	0.012		−0.005		−18.614	
Battery level	−0.001		0.002		−0.858	
Battery status (charging)	0.034		0.035		64.955	***
Battery status (discharging)	0.082	*	0.073	*	−26.642	
Location (Home)	0.157	***	0.149	***	−18.070	
Location (Other)	0.028		0.058		23.536	
Location (Transit)	0.152	***	0.147	***	−35.109	·
Location (Work)	0.125	*	0.134	**	−14.514	
Proximity (false)	−0.071	***	−0.070	***	23.332	**
Screen change time	−0.025	*	−0.019	*	−4.503	
Screen state (Off)	−0.030		−0.025		−0.720	
Screen on/off count	0.020	*	0.021	*	−2.447	
Weekend	−0.018		−0.002		−19.420	*
Morning message (8-10am)	−0.043	·	−0.019		24.587	*
Afternoon message (10am-6pm)	0.058	·	0.059	*	−59.742	***

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $\cdot\ p < 0.1$

S4 EXAMPLE OF INTERVENTION CONVERSATIONS

In this section we provide some examples of the actual conversations that took place between the coach (Walkie) and the participants. To maintain user privacy, the names in the conversations are fictitious.

The intervention messages were originally in German. We have translated some of them to English and include them here. It is important to note that we used Google Translate to translate the messages, and being a literal translation, they might not have the same effect or meaning as the original German ones.

We show full day conversations between Walkie and 3 participants – Andreas, Lisa and Sarah (Fig 4, 5 and 6, respectively). Two of these participants received the 'Self-monitoring' prompt (hence a total of 3 conversations in the day). Andreas did not receive the prompt (total of two conversations in the day).

To highlight variations with a single participants, we include examples of the Goal achievement (evening) message on three different days, in Fig. 7. The participant achieved the step goals on two days.

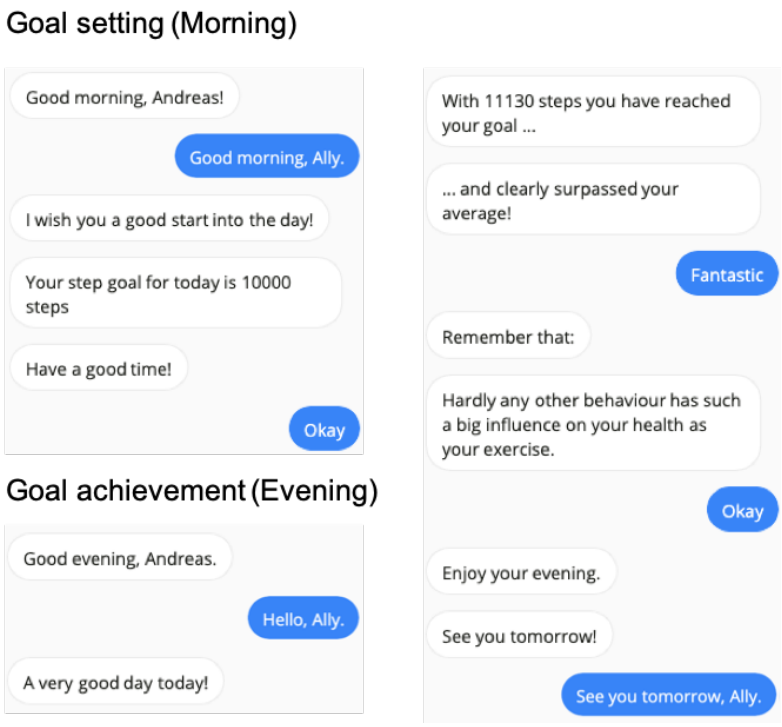


Fig. 4. Conversation with Andreas: Two initiating messages in the day

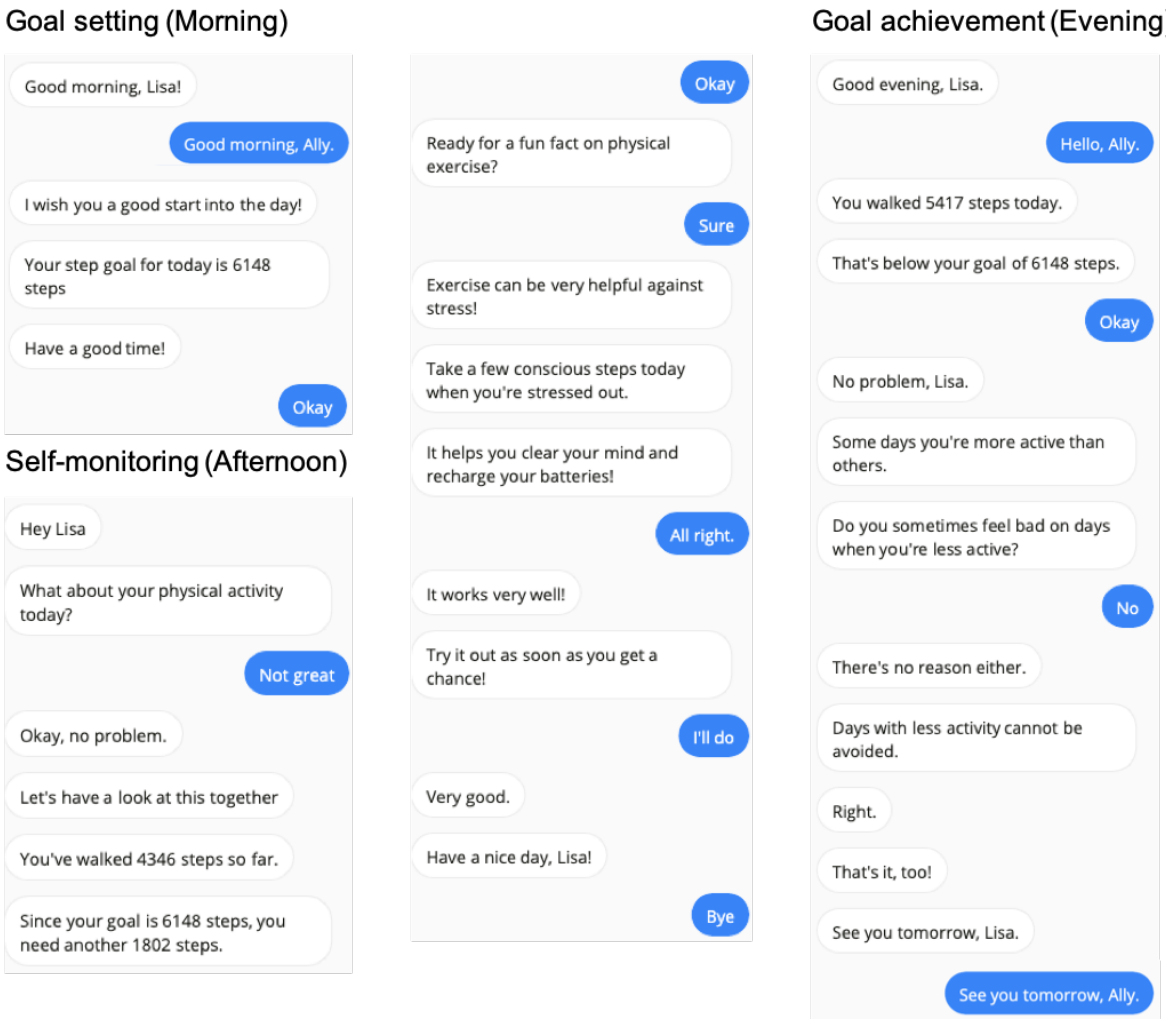


Fig. 5. Conversation with Lisa: Three initiating messages in the day

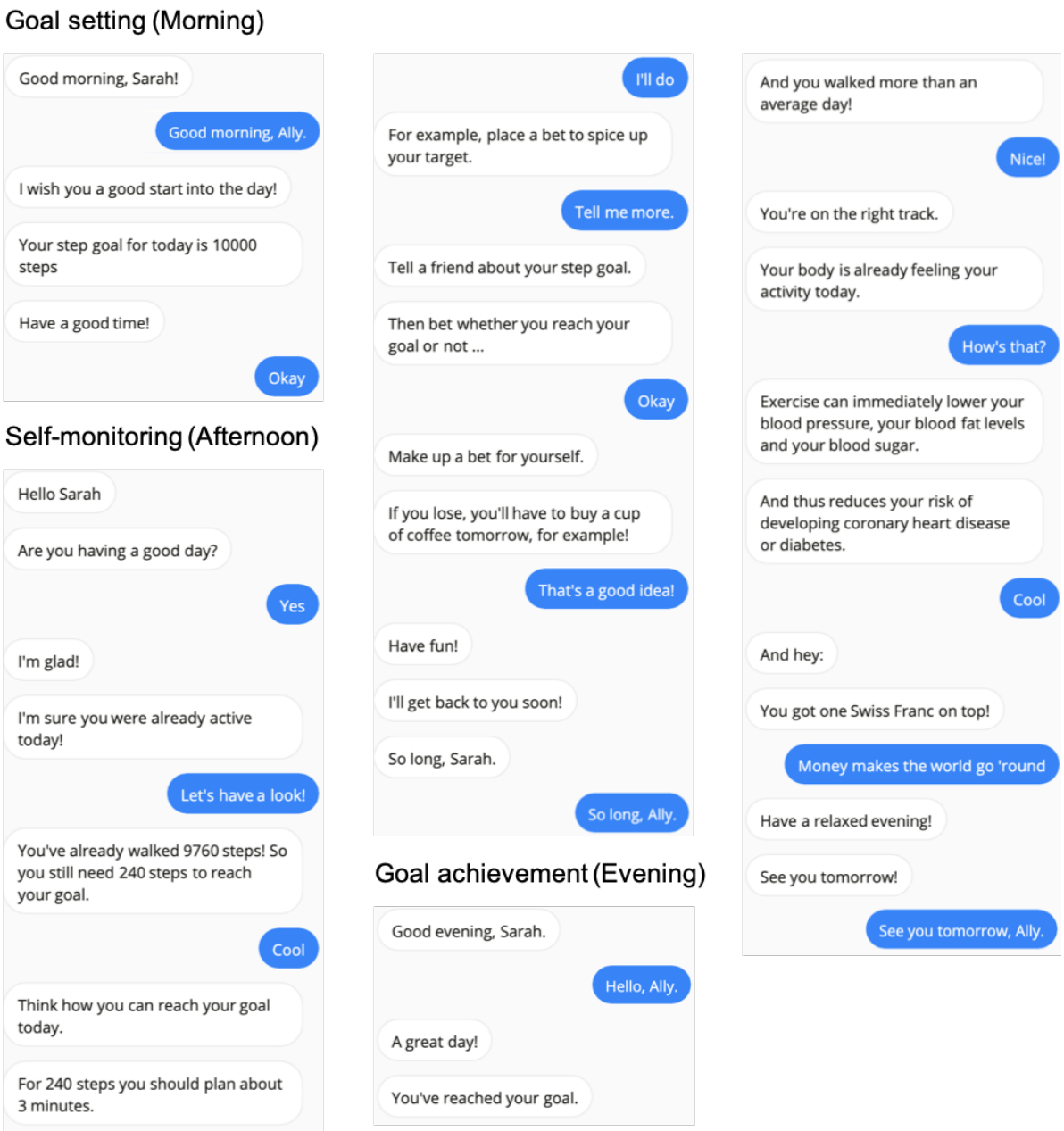


Fig. 6. Conversation with Sarah: Three initiating messages in the day

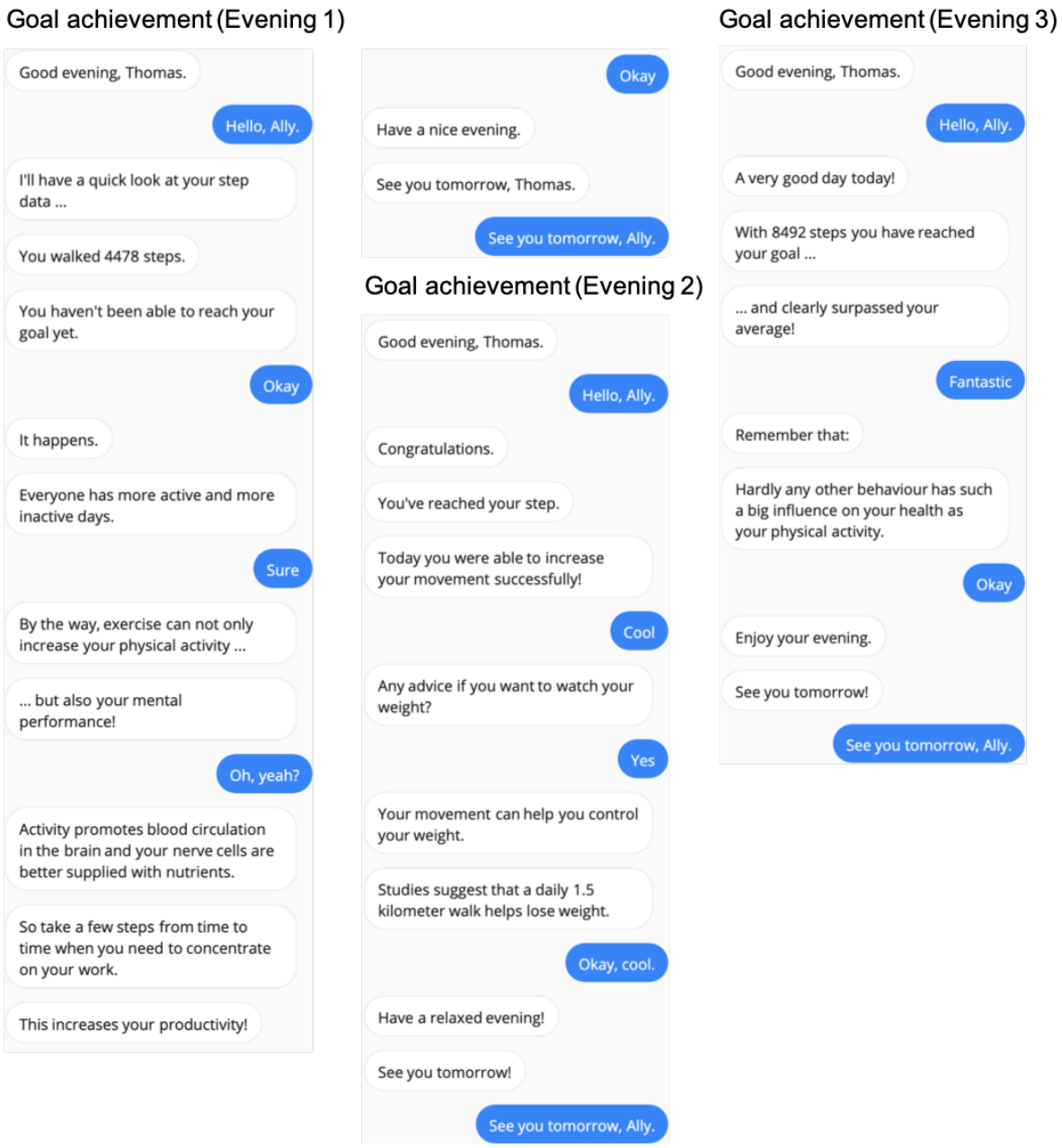


Fig. 7. Conversation with Thomas: Different goal achievement messages on three different days.

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